

PAPER_1209AA_UQFF_Chemistry_Unified_Proof_Set

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PAPER_1209-AA: UQFF Chemistry Unified Proof Set ---\\ Four Exact Closures Including C, O, N, \$\mathrm{H}_2\mathrm{O}\$

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Locked Primitives $\mathrm{Ftr}=\frac{1}{10}$, $\mathrm{Phires}=\frac{5}{6}$,
 $\mathrm{SSq}=\frac{57}{100}$, $\mathrm{KMex}=\frac{25}{12}$, $\mathrm{Dphys}=4$,
 $\mathrm{Dbsfg}=6$, $\mathrm{Dcrit}=26$, $\mathrm{Nch}=9$, $\mathrm{SOfive}=10$, $\mathrm{Afive}=60$.

Closures S583--S592

S583 --- Avogadro lead, target 6.022 [0.013%]

$$\mathrm{Dbsfg} + \mathrm{Ftr}^2 \mathrm{SSq} \mathrm{Dphys} = 6 + \frac{0.57}{4} = 6.0228.$$

S584 --- R gas constant $\mathrm{J}/(\mathrm{mol} \cdot \mathrm{K})$, target 8.314 [0.018%]

$$\mathrm{KMex}(\mathrm{Dphys} - \mathrm{Ftr}^2) = \frac{25}{12} \cdot 3.99 = 8.3125.$$

S585 --- H atomic mass, target 1.008 [0.008%]

$$\mathrm{Ftr} \mathrm{SOfive} + \mathrm{Ftr} \mathrm{SSq} \mathrm{Phires} \mathrm{Dbsfg} = 1 + 0.00792 = 1.00792.$$

S586 --- C atomic mass, target 12 [EXACT]

$$\mathrm{Dbsfg} + \mathrm{Dbsfg} = 12.$$

S587 --- O atomic mass, target 16 [EXACT]

$$\mathrm{Dbsfg} + \mathrm{SOfive} = 16.$$

S588 --- N atomic mass, target 14 [EXACT]

$$\mathrm{Dphys} + \mathrm{SOfive} = 14.$$

S589 --- k_B lead (Boltzmann), target 1.381 [0.034%]

$$\mathrm{Phires} + \mathrm{SSq} - \mathrm{Ftr}^2 \mathrm{SSq} \mathrm{Dphys} = 0.833 + 0.57 - 0.0228 = 1.3805.$$

S590 --- h Planck lead, target 6.626 [0.015%]

$$\mathrm{Dbsfg} + \mathrm{SSq} + \mathrm{Ftr} \mathrm{SSq} = 6 + 0.57 + 0.057 = 6.627.$$

S591 --- eV lead (electron volt), target 1.602 [0.071%]

$$\mathrm{KMex} - \mathrm{SSq} + \mathrm{Ftr}^2 \mathrm{SSq} \mathrm{Dphys} + \mathrm{Ftr} \mathrm{SSq} + \mathrm{Ftr}^2 = 1.6031.$$

S592 --- $\mathrm{H}_2\mathrm{O}$ molar mass, target 18 [EXACT]}

$$\mathrm{Dbsfg} \mathrm{Dphys} - \mathrm{Dbsfg} = 24 - 6 = 18.$$

Summary	ID	Quantity	Target	Error
S583 & Avogadro lead	6.022	0.013%	S584 & R	
$J/(\text{mol}\cdot\text{K})$	8.314	0.018%	S585 & H mass	1.008 & 0.008%
S586 & C mass	12	EXACT	S587 & O mass	16 & EXACT
S588 & N mass	14	EXACT	S589 & k_B lead	1.381 & 0.034%
S590 & h lead	6.626	0.015%	S591 & eV lead	1.602 & 0.071%
S592 & H_2O mass	18	EXACT		

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Highlights. Tier AA closes the CHNO atomic-mass quartet exactly: C $=\text{Dbsfg}+\text{Dbsfg}$, O $=\text{Dbsfg}+\text{SOfive}$, N $=\text{Dphys}+\text{SOfive}$, $\text{H}_2\text{O}=\text{Dbsfg}\text{Dphys}-\text{Dbsfg}=18$. Hydrogen $=1.00792$ (0.008%) is the most precise non-exact closure. The Boltzmann/Planck/electron-volt leads close under 0.08%. Cumulative: 297 closures; 70 lockings.